25AIMLPCC302ESE

15 OCT 2026

Seat No. : .....

Student Name : .....

St. John College of Engineering and Management, Palghar, Maharashtra

Autonomous Institute

End Semester Examinations, October 2026

B.Tech. Artificial Intelligence and Machine Learning -

SEMESTER III

25AIMLPCC302 - Data Structure

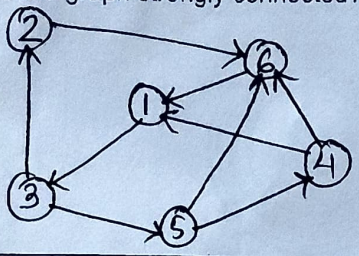
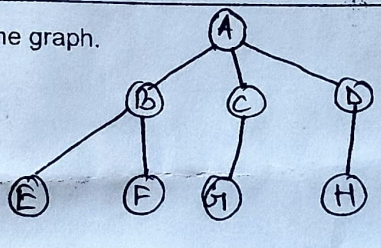
Time : 2.00 Hours **Maximum Marks : 60**

Instructions

- All Questions are compulsory
- Assume suitable data wherever required

St. John College of Engineering & Management SJCEM PALGHAR

		Attempt all Questions	Mrk	Course Outcome	RBT Level
1.	a	i. Explain Data structure? Identify and enlist few advantages of data structures	5	CO1	L3
		ii. What is Multi Dimenssional Array ? Give Example of 3-Dimensional Array. How the Elements are Accessed in 3-Dimenssional Array ?	5	CO1	L3
		OR			
	b	Explain the concept of an array of pointers. Compare an array of pointers with a two-dimensional array and discuss suitable applications of each.	10	CO1	L3
2.	a	i. Convert the given infix to Post Fix : a+b*c/d-e	5	CO2	L3
		ii. Explain Circular Queue With Example.	5	CO2	L3
		OR			
	b	Explain the ADT of a queue and describe enqueue and dequeue operations. Illustrate their array-based implementation with suitable algorithms.	10	CO2	L4
3.	a	i. Implement Queue : Delete Operation Using Linked List. [Pseudocode / Programme]	5	CO3	L3
		ii. Write the algorithm/pseudocode/program in any language to check whether the stack is empty or not. Provided Stack is implemented using Linked List	5	CO3	L3
		OR			
	b	Explain the different types of linked lists (singly, doubly, circular singly, circular doubly). Illustrate their node structure and typical use-cases for each.	10	CO3	L4

4.	a	i. Generate Huffman code for the letters a, b, c, d, e, f having frequencies 16, 8, 4, 2, 1, 1.	5	CO4	L3
		ii. Construct a BST from the given keys [25, 15, 50, 10, 22, 35, 70] and Perform inorder, preorder, and postorder traversals on the above tree.	5	CO4	L3
		OR			
	b	Insert the following elements into an AVL Tree: 21, 26, 31, 32, 35, 33, 34, 49, 64, 75 and show the tree after each rotation.	10	CO4	L3
5.	a	i. Draw a graph corresponding to the following adjacency matrix:1. Find indegree and outdegree of each vertex.2. Find the path matrix for this graph.3. Is the graph strongly connected? 4. Is the graph acyclic? 	5	CO5	L3
		ii. Perform DFS on the graph. 	5	CO5	L1
		OR			
	b	Explain the Depth First Search (DFS) algorithm and Breadth First Search (BFS) also compare between the two search algorithms	10	CO5	L3
6.	a	i. Compare Linear Probing, Quadratic Probing and Double Hashing.	5	CO6	L2
		ii. Explain linear search with example.	5	CO6	L2
		OR			
	b	Insert the following keys in an array of size 17 using the modulo division method. Use quadratic probing to resolve collisions: 94, 37, 29, 40, 84, 88, 102, 65, 74, 35.	10	CO6	L4

* All the Best *

St. John College of Engineering & Management ★ SJCEM PALGHAR